

Tabletop Manipulators Preliminary Design May 10:

Objective: Interact with table-mounted platform while leaving space for human interaction

→ 3 configurations may arise

- Human interacting with platform (needs room for human to interact comfortably)
- Human using platform, not interacting (keyboard/mouse)
- Robot only interaction (no constraints, robots must reach)

I) Human-platform interaction

→ Maximum width between human points of contact are size constraints for the platform, which dictates the minimum reach of the manipulators

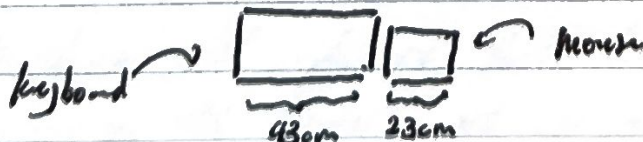
→ Assuming arms at shoulder width, ~60cm from elbow to elbow also rested on edge of table with hands in the middle of the table

→ No other constraints

$$|w > 60\text{cm}|$$

II) Human using platform freely

→ Maximum space occupied by keyboard + mouse is minimum width of platform



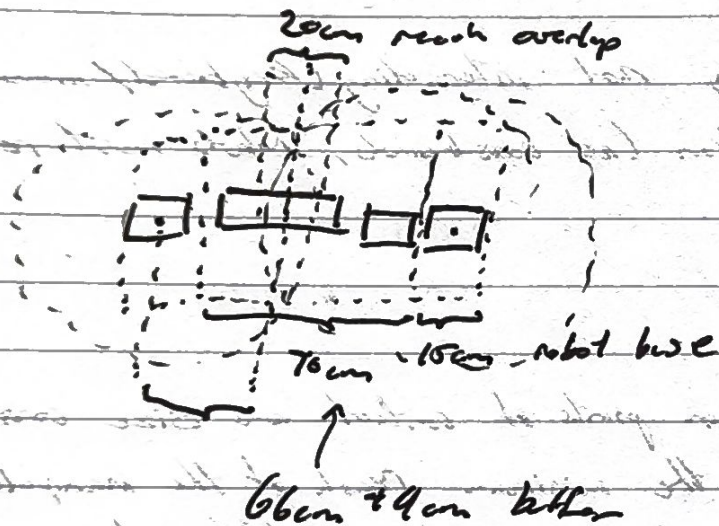
$$|w > 66\text{cm}|$$

III) Robot only interaction

→ Assuming robot base width of ~15cm, robot home ~25cm past width of platform

→ Now assume a minimum workspace overlap of 15-20cm

$$\begin{aligned} & \boxed{+15\text{cm to platform width workspace on top of platform width}} \\ & \boxed{+7.5-10\text{cm of reach per robot for overlap}} \end{aligned}$$

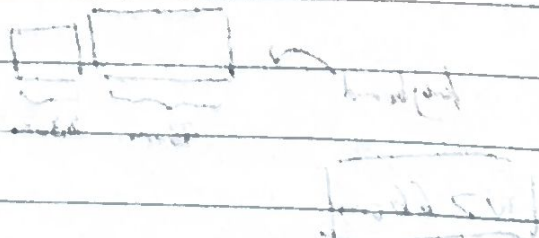


$$\begin{aligned} \text{Total reach} &= 35\text{cm} + 7.5\text{cm} + 10\text{cm} \Rightarrow 35\text{cm platform edge to} \\ &= 52.5\text{cm} \end{aligned}$$

⇒ 7.5cm robot base to platform edge
⇒ 10cm for workspace overlap

$$\rightarrow \boxed{52.5\text{cm total reach}}$$

→ 28cm shoulder to elbow } accounts for most of the reach
→ 20cm elbow to wrist
→ 15cm wrist to EE

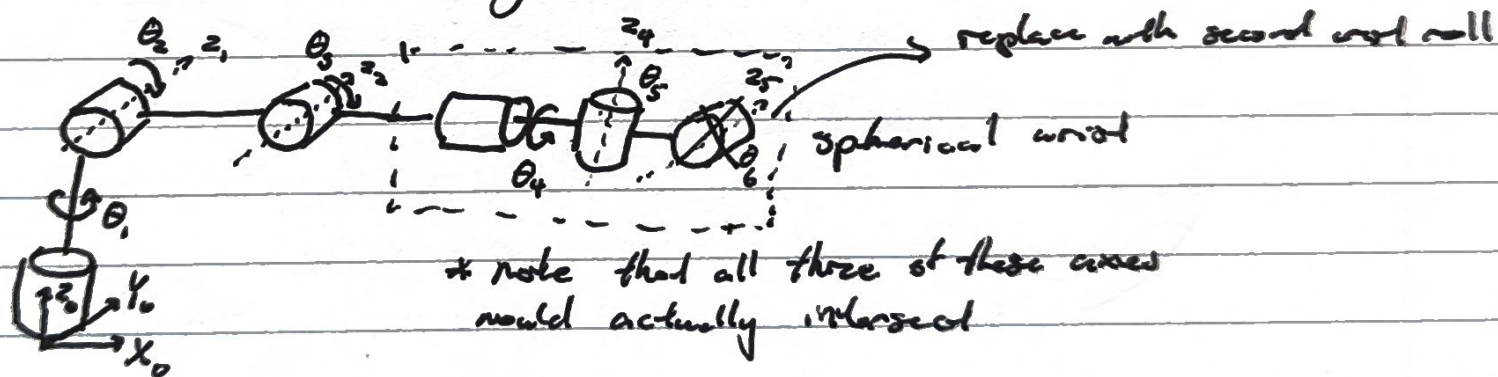


Finalized DH Parameters

May 25, 2026

The selected joint configuration for the arm is a GR arm with a spherical wrist. This configuration is originally proposed by Pieper and Roth in 1969 to reduce the number of potential configurations and the degree of the characteristic polynomial when finding an inverse kinematic solution. The principles and benefits of such a configuration hold true still and make this a good choice for this arm.

One added benefit of this configuration is the decoupling of the position and orientation of the end effector, also known as kinematic decoupling. The position of the wrist center is entirely determined by the base shoulder and elbow at any orientations can be achieved using only the joints of the spherical wrist, granted their joint axes intersect. The joints are illustrated below:



With then the 52.5 cm total reach, we assign DH parameters to define the kinematics of the arm. A couple things of note

- (i) Nearly, it not all of the reach will come from the shoulder-elbow and elbow-wrist links
- (ii) The base will sit at table level (potentially with actuation below) to ensure the base joint is at $z=0$

The parameters will be given in Modified DH convention, defined as:

- α_{i-1} (link twist) — angle between z_{i-1} and z_i measured about axis x_{i-1}
- a_{i-1} (link length) — distance between z_{i-1} and z_i measured along axis x_{i-1}
- d_i (link offset) — distance between x_{i-1} and x_i measured along z_i
- θ_i (link joint angle) — angle between x_{i-1} and x_i measured about axis z_i

The resulting DH table:

Joint (i)	α_{i-1}	a_{i-1}	d_i	θ_i
1 - base	0	0	0	θ_1
2 - shoulder	$-\frac{\pi}{2}$	0	0	θ_2
3 - elbow	0	270	0	θ_3
4 - wrist roll	$-\frac{\pi}{2}$	0	180	θ_4
5 - wrist pitch	$\frac{\pi}{2}$	0	0	θ_5
6 - wrist roll	$-\frac{\pi}{2}$	0	0	θ_6
7 - tool frame	0	0	75	0